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Department of Civil Engineering

IV B. Tech I Semester Question Bank

Subject: Environmental Impact Assessment **Year & Sem:** IV & I

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UNIT- IV

Assessment of Impact of development Activities on Vegetation and wildlife, Environmental Impact of Deforestation. Environmental Risk Assessment and Risk management in EIA: Risk assessment and treatment of uncertainty-key stages in performing an Environmental Risk Assessment- Advantages of Environmental Risk Assessment.

Q. No.	Question
1	Define Environmental Risk Assessment (ERA). **Environmental Risk Assessment (ERA)** is a systematic process used to identify environmental hazards, assess their likelihood and consequences, and estimate the potential risks to humans and ecosystems.
2	What is risk management? **Risk management** is the process of identifying, evaluating, controlling, and monitoring risks to reduce their harmful effects on the environment and human health.
3	Define hazard identification. **Hazard identification** is the process of identifying substances, activities, or conditions that have the potential to cause harm to humans, wildlife, vegetation, or the environment.
4	What is risk characterization? **Risk characterization** is the process of combining information on hazards, exposure, and their effects to estimate the nature and level of environmental risk.
5	Define uncertainty in risk assessment. **Uncertainty** is the lack of complete knowledge or information about hazards, exposure, environmental conditions, or effects that may affect the accuracy of risk estimates.
6	What are the impacts of developmental activities on vegetation? Developmental activities can cause **vegetation loss, habitat destruction, soil degradation, fragmentation of forests, reduced biodiversity, and changes in natural vegetation patterns**.
7	Define wildlife impact assessment. **Wildlife impact assessment** is the systematic evaluation of the effects of a proposed development activity on wildlife, their habitats, populations, and ecological relationships.
8	What is deforestation? **Deforestation** is the permanent removal or clearing of forest vegetation for purposes such as agriculture, construction, mining, industries, and infrastructure development.
9	State the environmental impacts of deforestation. Major impacts include **loss of biodiversity, soil erosion, habitat destruction, climate change, reduced rainfall, disturbance of the water cycle, and increased carbon emissions**.
10	What are the key stages in Environmental Risk Assessment? The key stages are **hazard identification, exposure assessment, effects assessment, risk characterization, uncertainty analysis, and risk evaluation**.

11		List the advantages of Environmental Risk Assessment. ERA helps to **identify environmental hazards, predict potential impacts, support decision-making, reduce environmental damage, manage uncertainty, and develop effective mitigation measures**.
12		What is ecological risk? **Ecological risk** is the probability that an environmental hazard or human activity will cause harmful effects on organisms, populations, communities, or ecosystems.
13		Define environmental hazard. An **environmental hazard** is any natural or human-induced substance, activity, or condition that can cause adverse effects on the environment, human health, wildlife, or ecosystems.
Q. No		**Question**
1	A	Explain the assessment of impacts of developmental activities on vegetation. **Assessment of Impacts of Developmental Activities on Vegetation** Developmental activities such as **urbanization, road construction, industries, mining, dams and agriculture** can adversely affect vegetation. The assessment generally involves: 1. **Baseline study** – Identify existing vegetation, forest types, species diversity and important plant species. 2. **Impact identification** – Assess vegetation clearing, tree loss, habitat destruction, fragmentation and pollution. 3. **Impact prediction** – Estimate the extent, magnitude, duration and significance of vegetation loss. 4. **Quantitative assessment** – Measure changes in vegetation cover, forest area, tree density and species diversity using **field surveys, GIS and Remote Sensing**. 5. **Mitigation measures** – Minimize vegetation clearing, undertake **afforestation, replantation, green-belt development and ecological restoration**. 6. **Monitoring** – Regularly monitor vegetation recovery and the effectiveness of mitigation measures.
	B	Discuss the assessment of impacts of developmental activities on wildlife. **Assessment of Impacts of Developmental Activities on Wildlife** Developmental activities such as **roads, industries, mining, dams, urbanization and agriculture** can adversely affect wildlife and their habitats. The assessment includes: 1. **Baseline study** – Identify wildlife species, their habitats, population and distribution in the project area. 2. **Impact identification** – Assess **habitat loss, habitat fragmentation, disturbance, noise, pollution and wildlife mortality**. 3. **Impact prediction** – Estimate the magnitude, extent, duration and significance of impacts on wildlife. 4. **Risk assessment** – Identify impacts on **endangered, rare and sensitive species** and evaluate ecological risks. 5. **Mitigation measures** – Provide wildlife corridors, habitat restoration, controlled construction activities and pollution/noise control. 6. **Monitoring** – Regularly monitor wildlife populations, habitat conditions and the effectiveness of mitigation measures.
2	A	Explain the causes and environmental impacts of deforestation. **Deforestation** is the permanent removal of forest cover for human activities. It is mainly caused by developmental and economic activities. **Causes of Deforestation** 1. **Agricultural expansion** – Forests are cleared for cultivation and plantations. 2. **Urbanization** – Forest land is converted for buildings and settlements. 3. **Industrial development** – Land is cleared for industries and infrastructure. 4. **Road and dam construction** – Large areas of forest may be cleared for projects.

		5. **Mining and quarrying** – Forest areas are removed for extraction of minerals. 6. **Fuelwood and timber collection** – Excessive cutting of trees reduces forest cover. 7. **Forest fires and overgrazing** – Damage and prevent natural regeneration. **Environmental Impacts** 1. **Loss of biodiversity** and wildlife habitat. 2. **Soil erosion** and loss of soil fertility. 3. **Climate change** due to increased carbon dioxide emissions. 4. **Disturbance of the water cycle** and reduced groundwater recharge. 5. **Increased floods and droughts** due to changes in runoff. 6. **Desertification and land degradation**. 7. **Loss of ecosystem services** such as carbon storage, climate regulation and habitat protection.
	B	Discuss measures for controlling and mitigating deforestation impacts. **Measures for Controlling and Mitigating Deforestation Impacts** Deforestation can be controlled by protecting existing forests and restoring degraded areas. Important measures include: 1. **Afforestation and reforestation** – Plant trees in degraded and cleared areas. 2. **Sustainable forest management** – Regulate tree cutting and promote controlled harvesting. 3. **Forest protection** – Prevent illegal logging, encroachment and uncontrolled forest fires. 4. **Social forestry** – Encourage tree planting in villages, roadsides and community lands. 5. **Agroforestry** – Combine trees with agricultural crops to reduce pressure on natural forests. 6. **Control of developmental activities** – Avoid unnecessary clearing of forests for roads, industries, mining and other projects. 7. **Wildlife habitat conservation** – Protect Forest corridors and ecologically sensitive areas. 8. **Environmental Impact Assessment (EIA)** – Assess forest impacts before approving major development projects. 9. **Public participation** – Involve local communities in forest protection and management. 10. **Regular monitoring** – Use **Remote Sensing and GIS** to monitor changes in forest cover.
3	A	Explain Environmental Risk Assessment and its importance in EIA. **Environmental Risk Assessment (ERA) and Its Importance in EIA** **Environmental Risk Assessment (ERA)** is a systematic process of identifying environmental hazards, estimating the likelihood and severity of their effects, and evaluating the risks to **human health, wildlife, vegetation and ecosystems**. **Main Steps of ERA** 1. **Hazard identification** – Identify activities or substances that may cause environmental harm. 2. **Exposure assessment** – Determine how humans or ecosystems may come into contact with the hazard. 3. **Effects assessment** – Study the possible harmful effects. 4. **Risk characterization** – Estimate the level and significance of risk. 5. **Risk management** – Develop measures to control or reduce the identified risks. 6. **Monitoring** – Monitor environmental conditions and effectiveness of mitigation measures. **Importance of ERA in EIA** • Identifies potential **environmental hazards and risks** before project implementation. • Helps predict the **severity and probability of adverse impacts**. • Supports better **project planning and decision-making**.

		• Helps develop suitable **mitigation and emergency measures**. • Protects **human health, biodiversity and natural resources**. • Helps address **uncertainty** in environmental predictions. • Ensures safer and more **sustainable development**.
	B	Discuss the objectives and principles of Environmental Risk Assessment. **Objectives and Principles of Environmental Risk Assessment** **Environmental Risk Assessment (ERA)** is a systematic process used to identify environmental hazards, evaluate their potential effects, and determine the level of risk to humans and ecosystems. **Objectives of ERA** 1. **Identify hazards** that may cause harm to the environment. 2. **Assess potential impacts** on human health, vegetation, wildlife, water, air and soil. 3. **Estimate the level of risk** based on the likelihood and severity of impacts. 4. **Identify vulnerable ecosystems and populations** that may be highly affected. 5. **Support decision-making** during environmental planning and project approval. 6. **Develop mitigation measures** to avoid or reduce environmental risks. 7. **Manage uncertainty** in environmental predictions. 8. **Promote sustainable development** by balancing development with environmental protection. **Principles of ERA** 1. **Scientific approach** – Risk assessment should be based on reliable scientific data. 2. **Precautionary principle** – Preventive action should be taken when serious environmental damage is possible. 3. **Systematic approach** – Risks should be assessed through clearly defined and organized steps. 4. **Transparency** – Assumptions, methods and results should be clearly documented. 5. **Risk-based decision-making** – Management actions should be based on the magnitude and significance of risk. 6. **Stakeholder participation** – Relevant stakeholders and local communities should be considered where appropriate. 7. **Continuous monitoring** – Risks and mitigation measures should be regularly reviewed.
4	A	Explain risk assessment and treatment of uncertainty in EIA studies. **Risk Assessment and Treatment of Uncertainty in EIA Studies** **Risk assessment** in Environmental Impact Assessment (EIA) is the process of identifying environmental hazards, estimating their probability and consequences, and determining the level of risk to **human health and the environment**. **Risk Assessment in EIA** The main steps are: 1. **Hazard identification** – Identify activities or substances that may cause environmental harm. 2. **Exposure assessment** – Determine the extent to which people, wildlife and ecosystems may be exposed. 3. **Impact assessment** – Evaluate the possible environmental effects. 4. **Risk characterization** – Estimate the overall level and significance of risk. 5. **Risk management** – Develop measures to avoid, reduce or control the risk. **Treatment of Uncertainty** **Uncertainty** occurs when complete information about environmental conditions, hazards, exposure or impacts is not available. It can be treated by: • Collecting **reliable and sufficient field data**. • Using appropriate **scientific models and methods**.

		• Considering different **possible scenarios**. • Performing **sensitivity and uncertainty analysis**. • Using conservative assumptions where necessary. • Clearly stating the **limitations and assumptions** of the assessment. • Conducting **regular environmental monitoring** and updating the assessment.
	B	Discuss different sources of uncertainty encountered in risk assessment. **Sources of Uncertainty in Risk Assessment** **Uncertainty** in risk assessment arises because complete and accurate information about environmental conditions, hazards and their effects is often unavailable. **Major Sources of Uncertainty** 1. **Data uncertainty** – Occurs due to insufficient, incomplete or inaccurate environmental data. 2. **Measurement uncertainty** – Results from limitations and errors in sampling and measurement instruments. 3. **Model uncertainty** – Arises because environmental models are simplified representations of complex natural systems. 4. **Parameter uncertainty** – Occurs when the exact values of model parameters are not known. 5. **Exposure uncertainty** – Results from variations in the amount, duration and pathways of exposure. 6. **Natural variability** – Environmental conditions naturally vary with **time, location and climate**. 7. **Future uncertainty** – Difficulty in predicting future land use, population growth, climate conditions and development activities. 8. **Scientific uncertainty** – Limited knowledge about the relationship between pollutants, ecological processes and environmental effects.
5	A	Explain the key stages involved in Environmental Risk Assessment. **Key Stages Involved in Environmental Risk Assessment (ERA)** **Environmental Risk Assessment (ERA)** is a systematic process used to identify hazards, evaluate their effects and determine the level of environmental risk. **Key Stages** 1. **Hazard Identification** Identify substances, activities or conditions that may cause harm to **human health, wildlife, vegetation and ecosystems**. 2. **Exposure Assessment** Determine **who or what is exposed, the exposure pathway, amount and duration** of exposure. 3. **Effects Assessment** Evaluate the potential harmful effects of the identified hazards on humans and the environment. 4. **Risk Characterization** Combine hazard, exposure and effects information to estimate the **probability and severity of adverse impacts**. 5. **Uncertainty Analysis** Identify uncertainties arising from **limited data, measurement errors, model assumptions and natural variability**. 6. **Risk Evaluation** Determine whether the estimated risk is **acceptable, tolerable or requires further control**. 7. **Risk Management and Monitoring** Develop **mitigation and control measures**, implement them, and monitor their

		effectiveness.
	B	Discuss hazard identification and risk analysis techniques used in EIA. **Hazard Identification and Risk Analysis Techniques Used in EIA** **Hazard identification** is the process of identifying activities, substances or conditions that may cause harm to **humans, wildlife, vegetation and the environment**. **Risk analysis** determines the likelihood and severity of these hazards. **1. Hazard Identification Techniques** • **Checklist method** – Uses a prepared list of possible environmental hazards. • **Hazard and Operability Study (HAZOP)** – Systematically identifies hazards and operational problems in a process. • **What-if analysis** – Examines possible consequences by asking *"What if?"* different failures or events occur. • **Fault Tree Analysis (FTA)** – Identifies possible causes leading to a major hazardous event. • **Failure Mode and Effects Analysis (FMEA)** – Examines possible equipment or process failures and their effects. **2. Risk Analysis Techniques** • **Qualitative risk analysis** – Classifies risks as **low, medium or high** based on likelihood and consequence. • **Quantitative risk analysis** – Uses numerical data to estimate the **probability and magnitude** of risks. • **Risk matrix** – Combines the **likelihood of occurrence and severity of consequences** to rank risks. • **Consequence analysis** – Determines the potential environmental effects of accidents or hazardous events. • **Scenario analysis** – Evaluates different possible accident or environmental-impact scenarios.
6	A	Explain risk characterization and risk evaluation procedures. **Risk Characterization and Risk Evaluation Procedures** **Risk characterization** is the process of combining information on **hazard, exposure and environmental effects** to estimate the level and significance of risk. **1. Risk Characterization Procedure** 1. **Combine assessment results** – Integrate hazard, exposure and effects information. 2. **Estimate risk** – Determine the **probability and severity** of adverse environmental effects. 3. **Identify affected receptors** – Assess impacts on **humans, wildlife, vegetation, water, soil and ecosystems**. 4. **Consider uncertainty** – Identify limitations in data, models and assumptions. 5. **Describe risk** – Classify the risk as **low, moderate or high** and explain its significance. **2. Risk Evaluation Procedure** Risk evaluation determines whether the identified risk is **acceptable or requires further action**. 1. Compare the estimated risk with **standards, guidelines or acceptable risk levels**. 2. Identify risks that require **control or mitigation**. 3. Consider the **severity, duration and reversibility** of impacts. 4. Compare alternative project options and mitigation measures. 5. Decide whether the project can proceed with suitable **risk-control measures**. 6. Establish **monitoring and review** requirements.
	B	Discuss various methods adopted for environmental risk management. **Methods of Environmental Risk Management** **Environmental Risk Management** is the process of identifying, controlling and reducing environmental risks associated with developmental activities.

		Major Methods 1. **Risk Avoidance** – Avoid activities or locations that may cause serious environmental damage. 2. **Risk Reduction** – Reduce the likelihood or severity of impacts through pollution-control and safer technologies. 3. **Risk Control** – Use engineering and administrative measures to control identified hazards. 4. **Mitigation Measures** – Adopt measures such as **waste treatment, emission control, afforestation and habitat restoration**. 5. **Emergency Preparedness** – Prepare emergency response plans for accidents, spills, fires and other hazardous events. 6. **Monitoring and Surveillance** – Regularly monitor **air, water, soil, vegetation and wildlife** to detect changes. 7. **Environmental Management Plan (EMP)** – Implement a planned programme for mitigation, monitoring and environmental protection. 8. **Risk Transfer/Sharing** – Share financial risks through **insurance and contractual arrangements**, where appropriate. 9. **Public Participation** – Involve local communities and stakeholders in identifying and managing environmental risks. 10. **Adaptive Management** – Review monitoring results and modify management measures when environmental conditions change.
7	A	Explain ecological risk assessment with suitable examples. **Ecological Risk Assessment with Examples** **Ecological Risk Assessment (ERA)** is a systematic process used to evaluate the likelihood that human activities, pollutants or natural hazards may cause harmful effects on **plants, animals, habitats and ecosystems**. **Main Steps** 1. **Problem formulation** – Identify the ecosystem, hazards and ecological components that may be affected. 2. **Exposure assessment** – Determine how organisms or ecosystems may be exposed to the hazard. 3. **Effects assessment** – Study the possible harmful effects on organisms, populations and habitats. 4. **Risk characterization** – Combine exposure and effects information to estimate the level of ecological risk. 5. **Risk management** – Develop suitable measures to reduce or control the identified risk. **Suitable Examples** • **Industrial wastewater:** Discharge of toxic chemicals into a river may affect fish and aquatic plants. • **Pesticide use:** Excessive pesticides may harm pollinators, soil organisms and other non-target species. • **Mining:** Mining activities may cause habitat destruction, soil contamination and loss of biodiversity. • **Road construction:** Roads through forest areas can cause habitat fragmentation and disturb wildlife movement. • **Oil spills:** Oil contamination can damage aquatic organisms, birds and coastal ecosystems.
	B	Discuss the significance of ecological risk assessment in environmental planning. **Significance of Ecological Risk Assessment in Environmental Planning** **Ecological Risk Assessment (ERA)** is a systematic process used to identify and evaluate the potential risks of developmental activities to **plants, animals, habitats and ecosystems**. It is

		important for planning environmentally sustainable projects. **Significance** 1. **Identifies ecological risks** – Detects potential threats to ecosystems before project implementation. 2. **Protects biodiversity** – Helps conserve **rare, endangered and sensitive species**. 3. **Supports project planning** – Helps select locations and designs with minimum ecological damage. 4. **Protects habitats** – Identifies important habitats, wildlife corridors and ecologically sensitive areas. 5. **Guides mitigation measures** – Helps develop measures such as **habitat restoration, afforestation and pollution control**. 6. **Supports EIA decision-making** – Provides scientific information for evaluating environmental impacts and project alternatives. 7. **Prevents long-term damage** – Helps identify risks before they become severe or irreversible. 8. **Promotes sustainable development** – Balances developmental requirements with **ecosystem conservation**. 9. **Improves environmental monitoring** – Establishes indicators for monitoring changes in ecosystem conditions. 10. **Supports regulatory decisions** – Helps authorities determine appropriate environmental safeguards and conditions.
8	A	Explain the role of Environmental Risk Assessment in sustainable development. **Role of Environmental Risk Assessment in Sustainable Development** **Environmental Risk Assessment (ERA)** is a systematic process of identifying, evaluating and managing environmental risks associated with developmental activities. It helps achieve development while minimizing damage to the environment. **Role of ERA** 1. **Identifies environmental risks** before implementing development projects. 2. **Protects natural resources** such as water, soil, forests and biodiversity. 3. **Supports sustainable project planning** by selecting safer alternatives and locations. 4. **Reduces pollution** through appropriate control and mitigation measures. 5. **Protects human health and ecosystems** from harmful environmental hazards. 6. **Promotes efficient resource use** and reduces unnecessary environmental losses. 7. **Supports EIA decision-making** by providing scientific information about potential risks. 8. **Addresses uncertainty** and helps prepare for unexpected environmental impacts. 9. **Encourages monitoring and adaptive management** throughout the project life cycle. 10. **Balances development and environmental protection**, which is essential for long-term sustainability.
	B	Discuss the relationship between risk assessment and environmental decision making. **Relationship Between Risk Assessment and Environmental Decision-Making** **Risk assessment** provides scientific information about potential environmental hazards and their consequences, while **environmental decision-making** uses this information to select suitable actions and alternatives. **Relationship** 1. **Hazard identification** – Identifies activities or substances that may cause environmental harm. 2. **Risk estimation** – Determines the **likelihood and severity** of possible impacts. 3. **Risk comparison** – Compares different project alternatives based on their environmental risks. 4. **Mitigation planning** – Helps decision-makers select appropriate measures to reduce

		risks. 5. **Priority setting** – High and unacceptable risks receive greater attention and control. 6. **Decision support** – Risk information helps authorities decide whether a project should be **approved, modified or rejected**. 7. **Monitoring and review** – Post-project monitoring provides information for revising decisions and management measures.
9	A	Explain the advantages of Environmental Risk Assessment in project planning. **Advantages of Environmental Risk Assessment in Project Planning** **Environmental Risk Assessment (ERA)** helps identify and evaluate potential environmental risks before a project is implemented. **Advantages** 1. **Early identification of risks** – Identifies environmental hazards at the planning stage. 2. **Better project selection** – Helps select suitable project locations, designs and alternatives. 3. **Prevents environmental damage** – Enables risks to be avoided or reduced before they occur. 4. **Improves decision-making** – Provides scientific information for project approval and planning. 5. **Supports mitigation planning** – Helps develop suitable measures to control environmental impacts. 6. **Protects natural resources** – Reduces impacts on **water, soil, forests, wildlife and biodiversity**. 7. **Reduces project delays and costs** – Early risk identification can prevent accidents, environmental violations and costly corrective actions. 8. **Improves public safety** – Helps protect human health and nearby communities. 9. **Supports legal compliance** – Assists in meeting environmental standards and regulatory requirements. 10. **Promotes sustainable development** – Balances development requirements with long-term environmental protection.
	B	Discuss the applications of Environmental Risk Assessment in industrial projects. **Applications of Environmental Risk Assessment in Industrial Projects** **Environmental Risk Assessment (ERA)** is used in industrial projects to identify, evaluate and manage risks that may affect **human health and the environment**. **Major Applications** 1. **Hazard identification** – Identifies hazardous chemicals, processes, equipment and activities. 2. **Pollution control** – Assesses risks from **air emissions, wastewater, solid waste and hazardous waste**. 3. **Chemical accident prevention** – Evaluates the possibility and consequences of **leaks, spills, fires and explosions**. 4. **Worker and public safety** – Helps reduce risks to workers and nearby communities. 5. **Site selection** – Helps identify locations with lower environmental sensitivity and risk. 6. **Environmental Impact Assessment (EIA)** – Provides risk information for evaluating proposed industrial activities. 7. **Emergency planning** – Supports preparation of emergency response and disaster management plans. 8. **Resource protection** – Assesses potential impacts on **soil, surface water, groundwater, vegetation and wildlife**. 9. **Risk mitigation** – Helps select suitable pollution-control technologies and preventive measures. 10. **Monitoring and compliance** – Supports continuous environmental monitoring and

		compliance with environmental regulations.
10	A	Describe the framework for environmental risk management in EIA. **Framework for Environmental Risk Management in EIA** The **Environmental Risk Management framework** in EIA is a systematic process for identifying, assessing, controlling and monitoring environmental risks associated with a proposed project. **Main Stages** 1. **Risk Identification** – Identify potential hazards and environmental impacts of the project. 2. **Risk Assessment** – Evaluate the **likelihood and severity** of identified risks. 3. **Risk Evaluation** – Determine whether the risks are **acceptable or require control**. 4. **Risk Control and Mitigation** – Develop measures to **avoid, minimize or control** environmental risks. 5. **Emergency Preparedness** – Prepare plans for accidents, spills, fires and other unexpected events. 6. **Implementation of EMP** – Include mitigation and monitoring measures in the **Environmental Management Plan (EMP)**. 7. **Monitoring** – Regularly monitor air, water, soil, vegetation, wildlife and other environmental conditions. 8. **Review and Improvement** – Evaluate monitoring results and modify management measures when necessary.
	B	Discuss case studies related to environmental risk assessment and management. **Case Studies Related to Environmental Risk Assessment and Management** Environmental Risk Assessment (ERA) is applied to real projects to identify hazards, assess their consequences and develop suitable management measures. **1. Bhopal Gas Tragedy – Industrial Risk** The **Bhopal gas disaster (1984)** involved the release of toxic methyl isocyanate (MIC) from a pesticide plant. Poor safety systems, inadequate emergency preparedness and ineffective risk management contributed to severe impacts on human health and the environment. **Risk management lessons:** • Proper hazard identification and process safety. • Regular maintenance and safety inspections. • Emergency preparedness and community warning systems. • Effective monitoring of hazardous substances. **2. Deepwater Horizon Oil Spill – Marine Risk** The **Deepwater Horizon accident (2010)** caused a major oil spill in the Gulf of Mexico. The spill affected marine organisms, coastal habitats, fisheries and local ecosystems. **Risk management measures:** • Improved offshore drilling safety. • Emergency spill-response planning. • Continuous monitoring of marine ecosystems. • Restoration of affected habitats. **3. Mining Projects – Ecological Risk** Mining can cause **deforestation, soil erosion, water pollution, habitat destruction and biodiversity loss**. ERA is used to identify sensitive ecosystems and predict potential impacts before mining begins. **Risk management measures:** • Controlled land clearing. • Proper waste and tailings management. • Water and soil pollution control. • Mine-site reclamation and afforestation.

11	A	Explain the impact assessment of developmental activities on vegetation and wildlife with suitable examples. **Impact Assessment of Developmental Activities on Vegetation and Wildlife** Developmental activities such as **roads, dams, mining, industries and urbanization** can cause significant changes to vegetation and wildlife. Impact assessment identifies these effects and helps develop suitable mitigation measures. **1. Impact on Vegetation** Developmental activities may cause: • **Removal of trees and forest cover** for construction. • **Habitat destruction and fragmentation**. • **Soil erosion and land degradation**. • Reduction in **plant species diversity**. • Dust and air pollution affecting plant growth. • Introduction of **invasive species**. **Example:** Construction of a highway through a forest may require tree cutting and can divide continuous forest into smaller vegetation patches. **2. Impact on Wildlife** Developmental activities may result in: • **Loss of wildlife habitat**. • Disturbance due to **noise, traffic and human activities**. • Restriction of animal movement. • Increased **wildlife mortality** due to road accidents. • Reduction in food and breeding areas. • Decline in biodiversity. **Example:** A road passing through a forest can interrupt wildlife movement and increase the risk of animals being killed by vehicles. **3. Assessment Methods** Impacts can be assessed through: • **Field surveys** of vegetation and wildlife. • **GIS and Remote Sensing** for mapping habitat and vegetation changes. • Species population and biodiversity studies. • Assessment of habitat loss and fragmentation. • Evaluation of the **magnitude, duration and significance** of impacts. **4. Mitigation Measures** • Minimize unnecessary vegetation clearing. • Implement **afforestation and habitat restoration**. • Provide **wildlife corridors and underpasses**. • Control construction noise, dust and pollution. • Protect endangered and sensitive species. • Conduct regular environmental monitoring.
	B	Discuss environmental hazards, risk mitigation measures and their importance in EIA studies. **Environmental Hazards, Risk Mitigation Measures and Their Importance in EIA** **Environmental hazards** are substances, activities or conditions that can cause adverse effects on **human health, vegetation, wildlife and ecosystems**. In EIA, identifying these hazards helps in predicting and controlling potential environmental risks. **1. Major Environmental Hazards** • **Air pollution** from industries, vehicles and construction. • **Water pollution** from industrial effluents and sewage. • **Soil contamination** due to hazardous waste and chemicals. • **Deforestation and habitat destruction**. • **Noise and vibration** from construction and industries.

- **Chemical spills, fires and explosions**.
- **Natural hazards** such as floods, droughts and landslides.

2. Risk Mitigation Measures

Risk mitigation measures are actions taken to **avoid, minimize or control** environmental risks.

1. **Avoidance** – Avoid environmentally sensitive areas and hazardous activities.
2. **Pollution control** – Install suitable air, water and waste-treatment systems.
3. **Vegetation protection** – Minimize tree cutting and undertake afforestation.
4. **Wildlife protection** – Provide wildlife corridors and protect important habitats.
5. **Safe waste management** – Properly collect, treat and dispose of hazardous waste.
6. **Emergency preparedness** – Develop plans for chemical spills, fires and accidents.
7. **Environmental monitoring** – Regularly monitor air, water, soil and ecological conditions.
8. **Environmental Management Plan (EMP)** – Include mitigation and monitoring measures in project implementation.

3. Importance in EIA

- Helps identify risks **before project implementation**.
- Reduces environmental damage and human health risks.
- Supports **project planning and decision-making**.
- Helps protect natural resources and biodiversity.
- Ensures compliance with environmental requirements.
- Reduces the probability and severity of environmental accidents.
- Promotes **sustainable and environmentally responsible development**.